Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Jr.Super60_NUCLEUS_BT	JEE-ADV-2023_P2	Date: 22-06-2025
Time: 02.00Pm to 05.00Pm	CTA-06	Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**D**	2	**B**	3	**C**	4	**C**	5	**ACD**	6	**ABC**
7	**AC**	8	**0**	9	**56**	10	**8**	11	**25**	12	**4**
13	**6**	14	**9**	15	**16**	16	**1**	17	**3**		

PHYSICS

18	**D**	19	**A**	20	**A**	21	**C**	22	**AB**	23	**AC**
24	**ACD**	25	**0**	26	**4**	27	**4**	28	**5**	29	**2**
30	**4**	31	**2**	32	**20**	33	**7**	34	**5.72 To 5.76**		

CHEMISTRY

35	**A**	36	**D**	37	**B**	38	**A**	39	**ABC**	40	**ABCD**
41	**ABD**	42	**3**	43	**2**	44	**1**	45	**20**	46	**84**
47	**3**	48	**6**	49	**19**	50	**1**	51	**2**		

SOLUTIONS

MATHEMATICS

1. Conceptual

2. $2\sin^2 x + 2\sin x.\cos^2 x + 4\sin x - 4 = 0$

$2\sin^2 x + 2\sin x.(1-\sin^2 x) + 4\sin x - 4 = 0$

$6\sin x - 4 = 0$

$\sin x = \frac{2}{3}$

$n = 5$ (in the given interval)

$x^2 + 5x + 2 = 0$

$x = \frac{-5 \pm \sqrt{17}}{2}$

Required interval $(-\infty, 0)$

3. $f(x) = 0 \Rightarrow \cos\left(\frac{n\pi}{x}\right).\cos(\pi x) = 1$

$\therefore \cos\left(\frac{n\pi}{x}\right) = 1$ and $\cos(\pi x) = 1$ (or) $\cos\left(\frac{n\pi}{x}\right) = -1$ & $\cos(\pi x) = -1$

$n = 33 \Rightarrow x = \pm 1, \pm 3, \pm 11, \pm 33$

No. of solutions = 8

4. $(\tan^2 x + 8\tan x + 15)(\tan^2 x + 8\tan x + 7) = 33$

Let $\tan^2 x + 8\tan x = \lambda$

Then $\lambda^2 + 22\lambda + 72 = 0 \quad \lambda = -18, -4 \quad \lambda = -18$ (rejected)

$\tan^2 x + 8\tan x + 4 = 0$

$\Rightarrow \tan x = -4 \pm \sqrt{12}$

5. $1 + \sum_{i=1}^{18} \lambda_i = 0 \quad \Rightarrow \sum_{i=1}^{18} \lambda_i = -1$

Squaring, $\sum_{i=1}^{18} \lambda_i^2 + 2 \sum_{1 \le i < j \le 18} \sum \lambda_i \lambda_j = 1$

$\Rightarrow \sum_{1 \le i < j \le 18} \sum \lambda_i \lambda_j = 1$

Now $\sum_{0 \le i < j < k \le 18} \lambda_i \lambda_j \lambda_k = 0$

$\Rightarrow \sum_{0 = i < j < k \le 18} \sum \sum \lambda_i \lambda_j \lambda_k + \sum_{1 \le i < j < k \le 18} \sum \sum \lambda_i \lambda_j \lambda_k = 0$

$\Rightarrow \sum_{1 \le i < j < k \le 18} \sum \sum \lambda_i \lambda_k + \sum_{1 \le i < j < k \le 18} \sum \sum \lambda_i \lambda_j \lambda_k = 0$

$\Rightarrow \sum_{1 \le i < j < k \le 18} \sum \sum \lambda_i \lambda_j \lambda_k = -1$

6. Simplify: $2\cos^4 x-1=0 \Rightarrow \cos^4 x=\frac{1}{2}$ and now proceed

7. $\sin^4 x+4\cos^4 x-4\sin^2 x\cos^2 x=0$

$\left(\sin^2 x-2\cos^2 x\right)=0$

$\tan^2 x=2$

$\cos 2\alpha=\frac{1-\tan^2 x}{1+\tan^2 x}=-\frac{1}{3}$

8. Conceptual

9. Conceptual

10. Conceptual

11. $\Rightarrow 2\cos 2\theta.\cos\frac{\theta}{2}=2\cos\frac{9\theta}{2}.\cos 3\theta$

$\Rightarrow \cos\frac{5\theta}{2}+\cos\frac{3\theta}{2}=\cos\frac{15\theta}{2}+\cos\frac{3\theta}{2}$

$\Rightarrow \cos\frac{15\theta}{2}=\cos\frac{5\theta}{2}$

$\Rightarrow \frac{15\theta}{2}=2k\pi\pm\frac{5\theta}{2}$

$5\theta=2k\pi$ or $10\theta=2k\pi$

$\theta=\frac{2k\pi}{5}$ $\quad\theta=\frac{k\pi}{5}$

$\therefore \theta=\left\{-\pi,\frac{-4\pi}{5},\frac{-3\pi}{5},\frac{-2\pi}{5},\frac{-\pi}{5},0,\frac{\pi}{5},\frac{2\pi}{5},\frac{3\pi}{5},\frac{4\pi}{5},\pi\right\}$

$m=5, n=5$

$\therefore mn=25$

12. $x^2+(2-\tan\theta)x-(1+\tan\theta)=0$

$\alpha+\beta=\tan\theta-2$ …. (1)

$\alpha\beta=-\tan\theta-1$ ….(2)

From equation (1) + (2)

$\alpha+\beta+\alpha\beta=-3$

$(\alpha+1)(\beta+1)=-2$

Hence either $\alpha+1=-2$ and $\beta+1=1$

Then $\alpha=-3,\beta=0$

or $\alpha+1=-1$ and $\beta+1=2$

$\alpha=-2$ and $\beta=1$

If $\alpha=-3,\beta=0$ then $\tan\theta=-1$

$\Rightarrow \alpha=\frac{3\pi}{4},\frac{7\pi}{4}$

If $\alpha=-2,\beta=1$ then $\tan\theta=1$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\text{Sum} = \frac{3\pi}{4} + \frac{7\pi}{4} + \frac{\pi}{4} + \frac{5\pi}{4} = 4\pi = 4$$

13. Conceptual

14-15. $\sin^2 3x = \dfrac{\sin^2 x}{\sin x - \frac{1}{4}}$

$$\Rightarrow \sin^2 3x - 1 = \frac{\left(\sin x - \frac{1}{2}\right)^2}{\left(\sin x - \frac{1}{4}\right)}$$

So $\sin x = \frac{1}{2}$ or 0

$$x = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, 2\pi, \frac{13\pi}{6}, \frac{17\pi}{6}, 3\pi, 4\pi$$

16-17.
$$\sin\left(1 + \sin^2 x + \sin^4 x\right) = \frac{13}{14}$$
$$\cos\left(1 + \sin^2 x + \sin^4 x\right) = \frac{-3\sqrt{3}}{14}$$

PHYSICS

18. The distance between them will be maximum when velocity of separation is zero and this condition will be fulfilled in this particular question when velocity of both particles are parallel.

19. Particle is undergoing uniform circular motion with centre at $(R, 0)$

20. Conceptual

21. The tangential acceleration of rope is $\dfrac{\lambda(4R)g}{2\lambda\pi R} = \dfrac{2g}{\pi}$.

Choose the upper quarter and apply NLM along the string, we get tension at right most point is zero.
Do same for upper half of string we get Tension at B is zero.
Do same for lower quarter of string we get tension at leftmost point is zero.

22. If friction were absent, then the acceleration of the load m would be greater than the acceleration of the load $3m$, which means that the friction force acting on the load m is directed to the left. At the moment of the beginning of slippage, a boundary situation arises: the maximum possible friction force acts in the system, but the accelerations of the loads are the same. Due to the weightlessness of the thread and the blocks, the tension force of the thread is equal to $\dfrac{F}{2}$

Let us write Newton's second law for the load m and load, $3m$ respectively:

$$\begin{cases} \frac{F}{2} - \mu mg = ma \\ F + \mu mg = 3ma \end{cases} \Rightarrow F = 8\mu mg$$

This means that there is no slippage at $F \le 8\mu mg$.

23. $2mg\sin\theta \le (\mu_1 + \mu_2)mg\cos\theta$

24. $T = \frac{mv^2}{\ell}$

25. Since there is no acceleration of cube in horizontal plane ($v = 5\hat{i} =$ constant) so no friction is required to move block along with cube.

26. By condition given

$R\sec\theta - R = R$, $\sec\theta = 2$, $\theta = 60^\circ$ $\tan\theta = \frac{x}{R}$ $v_0 = R\sec^2\theta\frac{d\theta}{dt}$ $v_0 = 4R\times\omega$

Hence $V_B = \frac{V_0}{4}$

27. Let F be the tension in the string. At the mass, the angle between the string and the radius of the dotted circle is $\theta = \sin^{-1}(r/R)$. In terms of θ, the radial and tangential $F = ma$ equations are

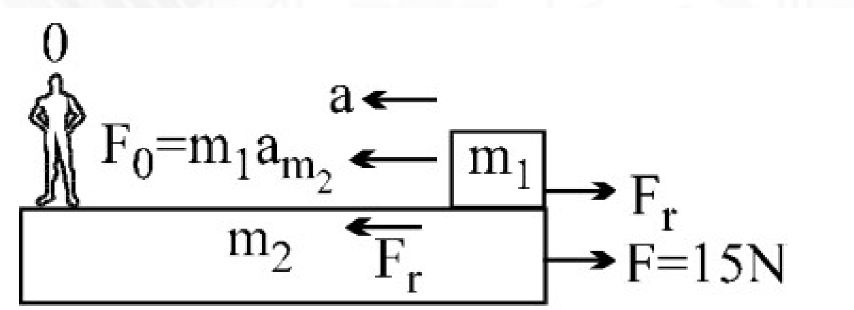
$F\cos\theta = \frac{mv^2}{R}$, and $F\sin\theta = m\dot{v}$

Dividing these two equations gives $\tan\theta = (R\dot{v})/v^2$. Separating variables and integrating gives

$$\int_{v_0}^{v}\frac{dv}{v^2} = \frac{\tan\theta}{R}\int_0^t dt \qquad � \frac{1}{v_0} - \frac{1}{v} = \frac{(\tan\theta)t}{R}$$

$$� v(t) = \left(\frac{1}{v_0} - \frac{(\tan\theta)t}{R}\right)^{-1}$$

28.



$Fr = \mu mg = 0.1\times1\times10 = 1\ N$

$a_{m_2} = \frac{F - Fr}{m_2} = \frac{15-1}{10} = 1.4\ m/sec^2$

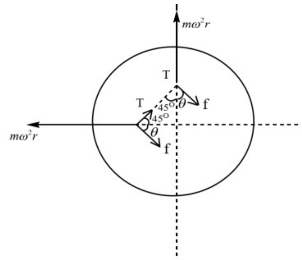
$a_{m_1, m_2} = a = \frac{m_1 a_{m_2} - Fr}{m_1} = \frac{1.4-1}{1}$

$= 0.4\ m/sec^2$

Time taken by m_1 to move 1 m

$t = \sqrt{\frac{2\times1}{0.4}} = \sqrt{5}$

29.



$$\theta = 45^\circ \; f_{mon} = \mu N - -(1)$$

$$f\cos\theta + T\cos45^\circ = mw^2r$$

$$f\sin\theta = T\sin45^\circ \text{ so, } mw^2r = f\left(\sin\theta + \cos\theta\right)$$

$$mw^2r = \mu m\sqrt{2}\left(\sin45^\circ\right) \text{ so } w_{mon} = \sqrt{\frac{\sqrt{2}\mu g}{r}}$$

30. Since the tape is weightless, the sum of the forces acting on it must be zero. This means that one or two bars should not move relative to the tape. Indeed, if both bars moved relative to the tape, then the difference in sliding friction forces would act on the tape in the direction along it

$\mu 2mg\cos\alpha - \mu mg\cos\alpha$

which is not equal to zero. It is obvious that the block of greater mass will be at rest relative to the tape, since if it were moving, the friction force acting from it on the tape would be equal to $2\mu mg\cos\alpha$, and the maximum force of friction at rest of the second body does not exceed $\mu mg\cos\alpha$,i.e. it cannot compensate for the first force.

$g\left(\sin\alpha - \frac{1}{2}\mu\cos\alpha\right)$

31 - 32.

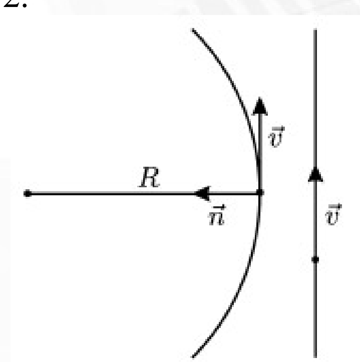


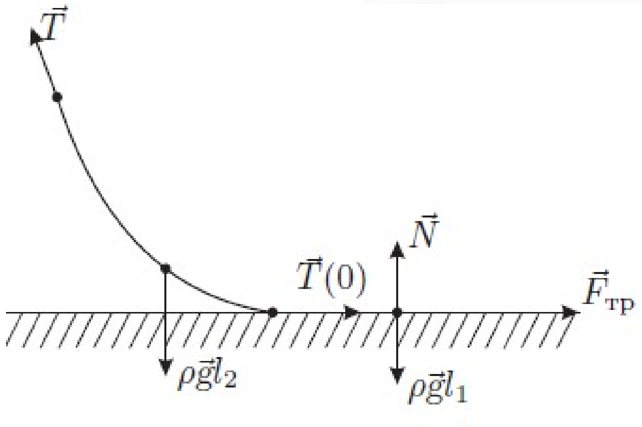
Fig. 1

Since at a certain moment of time t_0 the relative speed of cars is zero, the velocities of cars at this moment are equal both in magnitude and direction. Therefore, the values of the velocities of cars are the same. At the moment t_0 the acceleration of one car is zero, and the acceleration of the second car is equal to $v^2/R\,)\vec{n}$, where $\vec{n}$ is a unit vector in the direction to the center of the circle, on which this car is moving (see Fig. 1). Within a period of time $\Delta t = |t - t_0| \ll \frac{R}{v}$ The motion of the second car relative to the first car can be considered equally accelerated, so the relative velocity $\vec{V}_{!\backslash} \approx \frac{v^2}{R}\vec{n}\left(t - t_0\right)$, i.e. $\vec{v}_{\backslash} \approx \frac{v^2}{R}|t - t_0|$.

Drawing tangents to the graph near the time moment $t_0 = 20$ s (see Fig. 2), from their slope we obtain $\frac{v^2}{R} = \frac{\Delta v}{\Delta t} = \frac{40 \text{ m/s}}{20\text{s}} = 2 \text{ m/s}^2$,

$$v = \sqrt{\frac{\Delta v}{\Delta t} \cdot R} = \sqrt{2 \frac{\text{m}}{\text{s}^2} \cdot 200 \text{ m}} = 20 \frac{\text{m}}{\text{s}}.$$

33 - 34.



$$F = \lambda g \sqrt{H^2 + (kl)^2}$$
$$l_2 = \sqrt{H}\sqrt{(H + 2kl_1)}$$

CHEMISTRY

35. At constant $T, P,$ $\Delta G = -T\Delta S_{\text{total}} = \Delta H - T\Delta S$

$$\Rightarrow -400 \times (-40) = 2 \times 10^3 - 400\Delta S_{\text{Sys}}$$
$$\Rightarrow \Delta S_{\text{Sys}} = -35 J/\text{mol} K.$$

36. $\Delta S = nR\,\ell n\left(\frac{V_2}{V_1}\right) + nC_V\,\ell n\left(\frac{T_2}{T_1}\right)$

$$= R\,\ell n \frac{1}{2} + C_V\,\ell n 2$$
$$= (C_V - R)\ell n 2$$

37. $dG = VdP - SdT$

38. $\Delta G^\circ = 0 \Rightarrow T = 320K$

At $T > 320K, \Delta G_r^\circ < 1 \Rightarrow K_C > 1$

39. Process AB (isobaric)

$$\Delta S_{AB} = {}^{n}C_V \ell n\left(\frac{T_2}{T_1}\right)$$
$$= \frac{5R}{2}\ell n 2$$

Process BC:

$$\Delta S_{BC} = R\ell n \frac{10}{20} = -R\ell n 2$$
$$\Delta S_{Surroundings} = R\ell n 2$$

$= 2\ell n 1.5\,\text{Cal}\,\text{K}^{-1}$

$= 2(\ell n 3 - \ell n 2)\,\text{Cal}\,\text{K}^{-1}$

Process CA:

$\Delta S_{CA} = C_V \ell n \frac{300}{600}$

$= \frac{3R}{2} \ell n \frac{1}{2}$

$= -3\ell n 2\,\text{Cal}\,\text{K}^{-1}$

40. $\Delta H = -ve$ (given)

$\Delta S = +ve$ ($\because$ Gases are formed from solid).

$\therefore$ Reaction is spontaneous at all T.

As $T\uparrow$ spontaneously $\downarrow$ for exothermic reaction.

41. CONCEPTUAL

42. $\underset{217K}{H_2O(s)} \xrightarrow{\Delta S_1} \underset{273}{H_2O(s)} \xrightarrow{\Delta S_2} \underset{273K}{H_2O(l)} \xrightarrow{\Delta S_3} \underset{373K}{H_2O(g)}$

$\Delta S_1 = mS_1 \ell n \frac{273}{217} = 1 \times 2.1 \times 0.23 = 0.483\,\text{kJ}\,\text{K}^{-1}$

$\Delta S_2 = \frac{\Delta H_{vap}}{273} = \frac{344}{273} = 1.26\,\text{kJ}\,\text{K}^{-1}$

$\Delta S_3 = mS_2 \ell n \left(\frac{373}{273}\right) = 1 \times 4.2 \times 0.3 = 1.26\,\text{kJ}\,\text{K}^{-1}$

$\Delta S_{total} = 0.483 + 1.26 + 1.26 = 3.003\,\text{kJ}\,\text{K}^{-1}$

43. $\Delta H = \Delta U + \Delta n_g RT$

$= 2.8 \times 2 \times 2 \times 300 \times 10^{-3}$

$= 4.0\,\text{k cal}$

$\Delta G = \Delta H - T\Delta S$

$= 4 - 300 \times 20 \times 10^{-3}$

$= 4 - 6 = -2\,\text{k cal}$

44. $Q = -W = P_{ext}\Delta V$

$= -3\,L\,\text{bar} = -300J$

$Q_{surr} = 300J$

$\Delta S_{surr} = \frac{Q_{surr}}{T_{surr}} = \frac{300}{300} = 1\text{JK}^{-1}$

45. $T_{\min} = \frac{\Delta H_r^0}{\Delta S_r^0} = \frac{1.6 \times 10^3}{80} = 20K.$

46. $\Delta G = \int VdP - \int SdT$

$= \int_{P_1}^{P_2} \frac{nRT}{P} dP \qquad = nRT \ell n \left(\frac{P_2}{P_1}\right)$

$$= nRT\ell n\left(\frac{V_1}{V_2}\right)$$
$$= 0.1 \times 2 \times 300 \ell n 4$$
$$= 84\,\text{Cal}$$

47. $III, IV \& V$

$T_B = T_C = 300\,K$

$$Q_{BA} = {}^{n}C_{V_m}(T_A - T_B)$$
$$\Rightarrow\ 3000 = 10 \times \frac{3R}{2}(T_A - T_B)$$
$$\Rightarrow\ T_A = 4000\,K$$
$$\Delta U_{A \to D} = {}^{n}C_{V_m}(T_D - T_A)$$
$$\Rightarrow\ 3000 = 10 \times \frac{3R}{2}(T_D - T_A)$$
$$\Rightarrow\ T_D = 500\,K$$
$$W_{CD} = {}^{n}C_{V_m}\Delta T \text{ (Rev. Adiabatic Process)}$$
$$= 10 \times \frac{3R}{2}(T_D - T_C)$$
$$= 6000\,cal.$$

49. $\Delta S_{BC} = \Delta S_{BA} + \Delta S_{AD} + \Delta S_{DC}$

$$= {}^{n}C_{V_m} \ln\frac{T_A}{T_B} + {}^{n}C_{P_m} \ln\frac{T_D}{T_4} + 0$$
$$= 10 \times \frac{3R}{2} \ln\frac{400}{300} + 10 \times \frac{5R}{2} \ln\frac{500}{400}$$
$$= 19\,cal.K^{-1}.$$

50. I

51. IV, V